TIANSAWANG SUETRONG

ICT STUDENT

SOFTWARE ENGINEERING | DATA | IoT

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Portfolio tiansawang1818-pixel.github.io **GitHub** github.com/tiansawang1818-pixel

PROFILE

Digital Science and Technology undergraduate at Mahidol University seeking an internship in software engineering, data, or IoT. Built and deployed two full-stack systems that are live today: a no-code bot platform on Rust WebAssembly and a geospatial decision platform for Bangkok's urban heat, running on real satellite data.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL
Web and APIs: React, Next.js, FastAPI, Node.js, HTML, CSS, Tailwind CSS, REST APIs
Databases: PostgreSQL, PostGIS, MySQL, Redis, Supabase
Data and Modeling: ER modeling, relational database design, data validation, geospatial data, Google Earth Engine, OR-Tools
Tools: Git, GitHub, GitHub Actions, Docker, VS Code

PROJECTS

HEAT//SHIFT - Live Pilot

Python | FastAPI | Next.js | PostgreSQL/PostGIS | Redis/Celery | MapLibre | *GitHub Repository* | *Live Website*

- Built and deployed a map-first platform for planning heat mitigation investment across Bangkok's 50 districts.
- Designed a six-stage pipeline (staging, validation, processing, publishing, cache invalidation, live client updates) that monitors seven real sources, including Landsat 8/9, Sentinel-2, Dynamic World, WorldPop, and OpenStreetMap.
- Built versioned PostGIS datasets, a budget optimizer with Google OR-Tools, and evidence reports showing data versions, uncertainty, and stale-result warnings; admin tools are protected by OIDC single sign-on.
- Runs on a $0 cloud budget: four supervised processes in 768 MiB of RAM, with CI checks enforcing the same limits.

WasmBot Studio

TypeScript | React | Supabase | PostgreSQL | Rust WebAssembly | *GitHub Repository* | *Live Website*

- Built a no-code platform for creating rule-based business bots with a visual AND/OR editor supporting 12 operators.
- Designed versioned deployments with immutable snapshots and secured data using Supabase Auth, PostgreSQL row-level security, scoped API keys, and rate limiting.
- Integrated a Rust WebAssembly rule engine with seven Supabase Edge Functions and validated the deployed system through automated and end-to-end testing.

EDUCATION

Mahidol University | **2025 - Present**
Faculty of Information and Communication Technology
Bachelor of Science in Digital Science and Technology | GPA 3.08

Klaeng Wittayasathaworn School | **2022 - 2025**
Secondary Education

EXPERIENCE AND ACTIVITIES

MUICT Open House - Guest Relations | Mahidol University | 2026
Raknong G2 - Staff | Mahidol University | 2026

LANGUAGES

Thai: Native | English: Intermediate